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Lab Assignment 14.5

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|------------------|----------------------|
| Program          | : B. Tech (CSE)      |
| Specialization   | : -                  |
| Course Title     | : AI Assisted Coding |
| Course Code      | : 23CS002PC304       |
| Semester         | :II                  |
| Academic Session | : 2025-2026          |
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| Batch No :       | : 52                 |
| Date             | : 13/03/26           |

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### Task 1 – Image Gallery Website

Prompt:

Create an image gallery web page.

Requirements:

- Display images in a grid layout.
- Use AI to:
  - Optimize layout using CSS Grid.
  - Add hover zoom or overlay effects.
  - Ensure mobile responsiveness.

Code:

```

<? index.html > ...
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4    <meta charset="UTF-8">
5    <meta name="viewport" content="width=device-width, initial-scale=1.0">
6    <title>Image Gallery</title>
7    <link rel="stylesheet" href="styles.css">
8  </head>
9  <body>
10   <h1>Image Gallery</h1>
11   <div class="gallery">
12     <div class="gallery-item">
13       
14       <div class="overlay">
15         <p>Image 1</p>
16       </div>
17     </div>
18     <div class="gallery-item">
19       
20       <div class="overlay">
21         <p>Image 2</p>
22       </div>
23     </div>
24     <div class="gallery-item">
25       
26       <div class="overlay">
27         <p>Image 3</p>
28       </div>
29     </div>
30     <div class="gallery-item">
31       
32       <div class="overlay">
33         <p>Image 4</p>
34       </div>
35     </div>
36     <div class="gallery-item">
37       
38       <div class="overlay">
39         <p>Image 5</p>
40       </div>
41     </div>
42     <div class="gallery-item">
43       
44       <div class="overlay">
45         <p>Image 6</p>

```

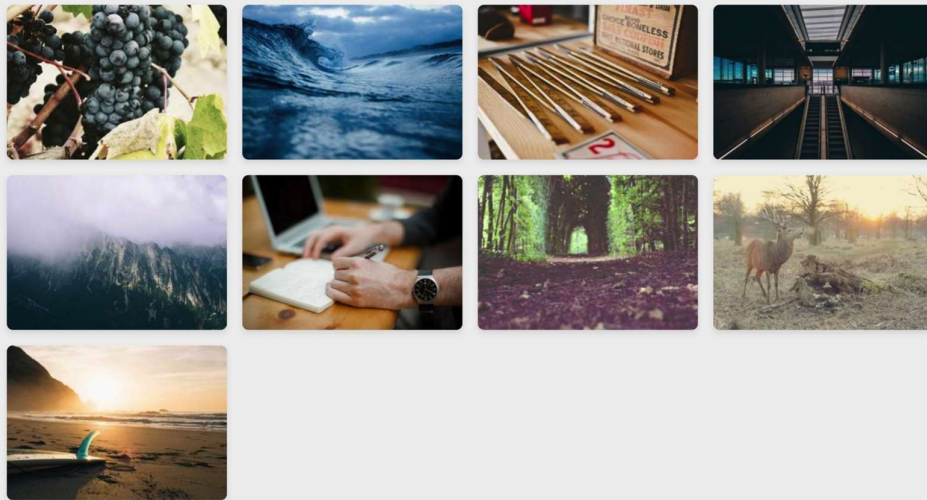
```

46       </div>
47     </div>
48     <div class="gallery-item">
49       
50       <div class="overlay">
51         <p>Image 7</p>
52       </div>
53     </div>
54     <div class="gallery-item">
55       
56       <div class="overlay">
57         <p>Image 8</p>
58       </div>
59     </div>
60     <div class="gallery-item">
61       
62       <div class="overlay">
63         <p>Image 9</p>
64       </div>
65     </div>
66   </div>
67 </body>
68 </html>

```

Output:

### Image Gallery



## Task 2 – Profile Card Generator

Prompt: Create a profile card web page.

### Requirements:

- Display user image, name, role, and social media links.
- Use AI to:
  - Suggest modern UI colors and fonts.
  - Design the card using Flexbox.
  - Add hover animation and shadow effects.

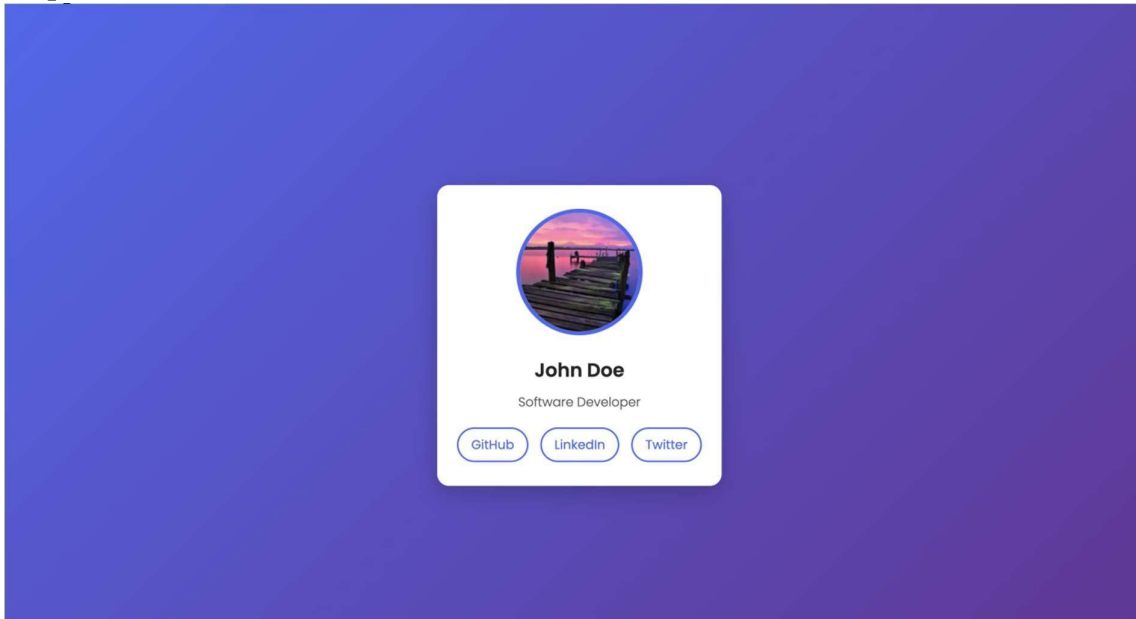
Code:

```

<> profile.html > html
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8">
5      <meta name="viewport" content="width=device-width, initial-scale=1.0">
6      <title>Profile Card</title>
7      <link href="https://fonts.googleapis.com/css2?family=Poppins:wght@300;400;600&display=block" rel="stylesheet">
8      <link rel="stylesheet" href="profile.css">
9  </head>
10 <body>
11     <div class="profile-card">
12         <div class="profile-image">
13             
14         </div>
15         <div class="profile-info">
16             <h1 class="name">Kowshik</h1>
17             <p class="role">Software Developer</p>
18             <div class="social-links">
19                 <a href="#" class="social-link" data-platform="github">GitHub</a>
20                 <a href="#" class="social-link" data-platform="linkedin">LinkedIn</a>
21                 <a href="#" class="social-link" data-platform="twitter">Twitter</a>
22             </div>
23         </div>
24     </div>
25 </body>
26 </html>

```

Output:



Task 3 -Develop a dynamic shopping cart system.

Prompt: Create a shopping cart system.

Requirements:

- Add, remove, and update product quantities.
- Display real-time total price.
- Use AI to:

- Manage cart state using JavaScript objects.
- Persist cart data in localStorage.

Code:

```

1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4    <meta charset="UTF-8">
5    <meta name="viewport" content="width=device-width, initial-scale=1.0">
6    <title>Shopping Cart</title>
7    <link rel="stylesheet" href="cart.css">
8  </head>
9  <body>
10   <h1>Shopping Cart System</h1>
11   <div class="container">
12     <div class="products">
13       <h2>Products</h2>
14       <div class="product-list">
15         <div class="product" data-id="1">
16           
17           <h3>Laptop</h3>
18           <p>$999.99</p>
19           <button class="add-to-cart">Add to Cart</button>
20         </div>
21         <div class="product" data-id="2">
22           
23           <h3>Mouse</h3>
24           <p>$29.99</p>
25           <button class="add-to-cart">Add to Cart</button>
26         </div>
27         <div class="product" data-id="3">
28           
29           <h3>Keyboard</h3>
30           <p>$79.99</p>
31           <button class="add-to-cart">Add to Cart</button>
32         </div>
33         <div class="product" data-id="4">
34           
35           <h3>Monitor</h3>
36           <p>$299.99</p>
37           <button class="add-to-cart">Add to Cart</button>
38         </div>
39       </div>
40     </div>
41     <div class="cart">
42       <h2>Cart</h2>
43       <div class="cart-items"></div>
44       <div class="cart-total">

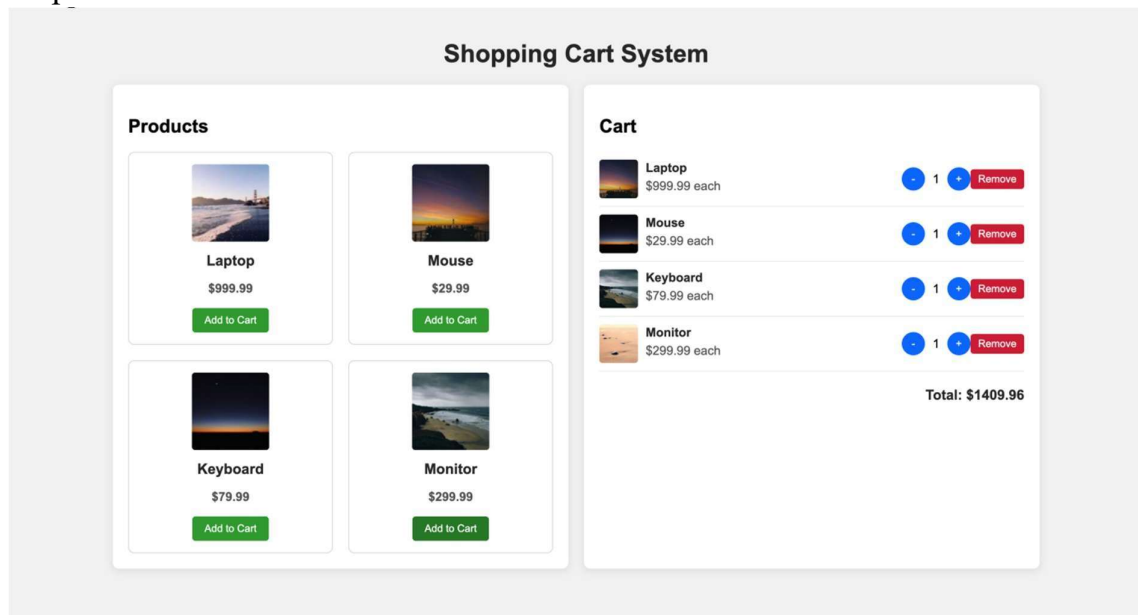
```

```

45         <strong>Total: $<span id="total">0.00</span></strong>
46       </div>
47     </div>
48   </div>
49   <script src="cart.js"></script>
50 </body>
51 </html>

```

Output:



Prompt: Create a role-based web application.

### Requirements:

- Admin, Editor, User roles.
- Conditional UI rendering.
- Use AI to:
  - Design access control logic.
  - Maintain secure state transitions.
  - Create scalable UI components.

### Code:

```

1  <!DOCTYPE html> ...
2  <html lang="en">
3  <head>
4    <meta charset="UTF-8">
5    <meta name="viewport" content="width=device-width, initial-scale=1.0">
6    <title>Role-Based Access Control</title>
7    <link rel="stylesheet" href="rbac.css">
8  </head>
9  <body>
10   <div id="login" class="login-container">
11     <h1>Login</h1>
12     <select id="role-select">
13       <option value="">Select Role</option>
14       <option value="user">User</option>
15       <option value="editor">Editor</option>
16       <option value="admin">Admin</option>
17     </select>
18     <button id="login-btn">Login</button>
19   </div>
20
21   <div id="dashboard" class="dashboard" style="display: none;">
22     <header class="header">
23       <h1>Dashboard</h1>
24       <div class="user-info">
25         <span id="current-role">Role: </span>
26         <button id="logout-btn">Logout</button>
27       </div>
28     </header>
29
30     <nav class="nav">
31       <button class="nav-btn" data-section="content">Content</button>
32       <button class="nav-btn" data-section="users" id="users-nav">Users</button>
33       <button class="nav-btn" data-section="settings" id="settings-nav">Settings</b
34     </nav>
35
36     <main class="main">
37       <section id="content-section" class="section">
38         <h2>Content Management</h2>
39         <p>Welcome to the content area.</p>
40         <button id="view-content-btn" class="action-btn">View Content</button>
41         <button id="edit-content-btn" class="action-btn">Edit Content</button>
42         <button id="delete-content-btn" class="action-btn">Delete Content</button>
43       </section>
44

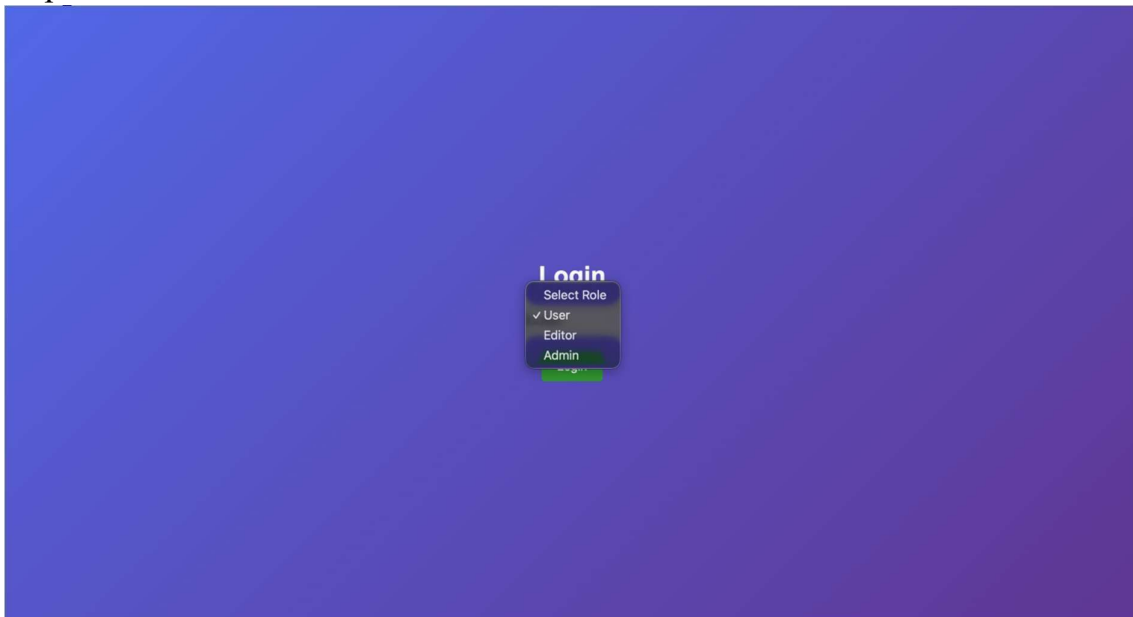
```

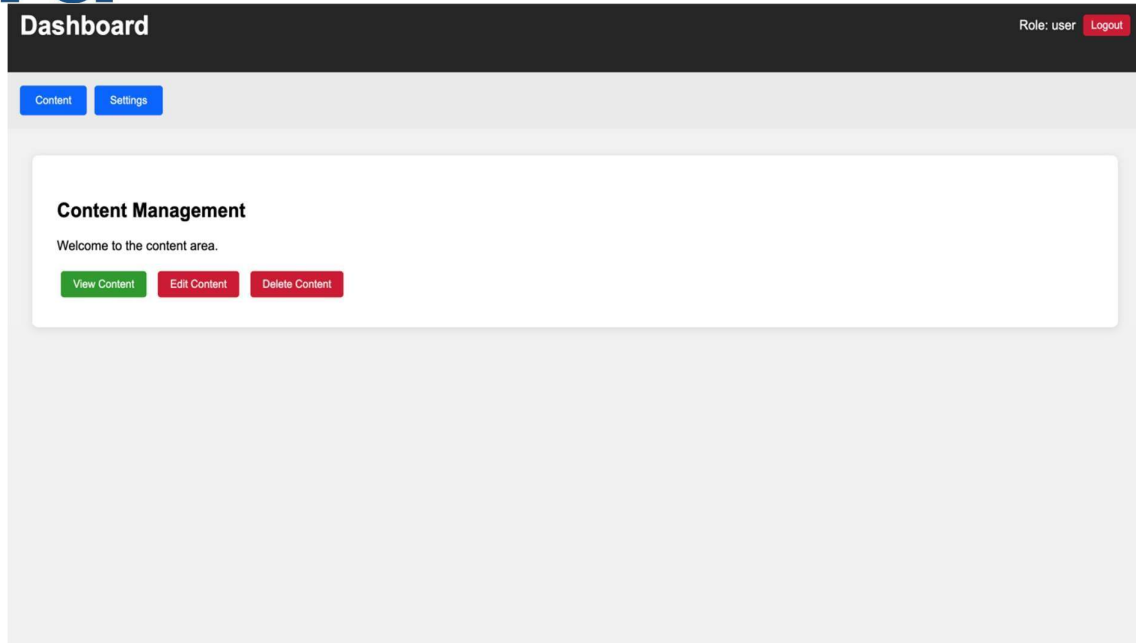
```

45     <section id="users-section" class="section" style="display: none;">
46         <h2>User Management</h2>
47         <p>Manage users here.</p>
48         <button id="view-users-btn" class="action-btn">View Users</button>
49         <button id="add-user-btn" class="action-btn">Add User</button>
50         <button id="delete-user-btn" class="action-btn">Delete User</button>
51     </section>
52
53     <section id="settings-section" class="section" style="display: none;">
54         <h2>Settings</h2>
55         <p>Application settings.</p>
56         <button id="view-settings-btn" class="action-btn">View Settings</button>
57         <button id="edit-settings-btn" class="action-btn">Edit Settings</button>
58     </section>
59 </main>
60 </div>
61
62     <script src="rbac.js"></script>
63 </body>
64 </html>

```

Output:





## Task 5 – Multi-Language Website

Prompt: Design a multilingual website.

Requirements:

- Switch languages dynamically.
- Persist language preference.
- Use AI to:
  - Structure translation files.
  - Handle RTL/LTR layouts.
  - Improve accessibility.



Code:

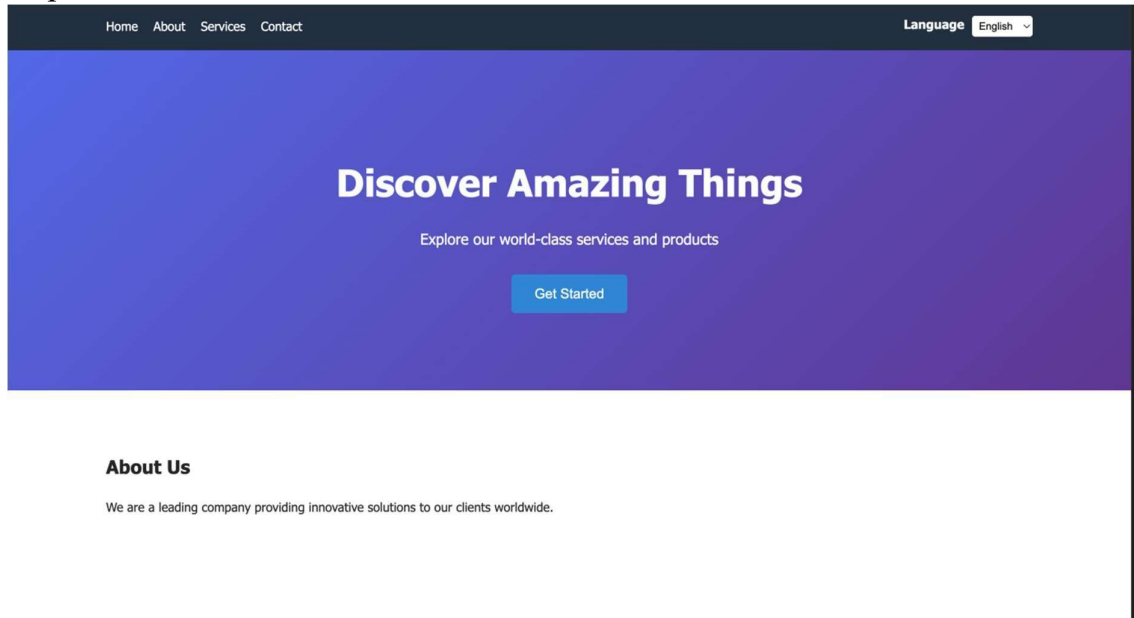
```

1  <!DOCTYPE html>
2  <html lang="en" dir="ltr">
3  <head>
4      <meta charset="UTF-8">
5      <meta name="viewport" content="width=device-width, initial-scale=1.0">
6      <title data-i18n="title">Welcome to Our Website</title>
7      <link rel="stylesheet" href="multi.css">
8  </head>
9  <body>
10     <header>
11         <nav>
12             <div class="nav-container">
13                 <ul>
14                     <li><a href="#" data-i18n="nav.home">Home</a></li>
15                     <li><a href="#" data-i18n="nav.about">About</a></li>
16                     <li><a href="#" data-i18n="nav.services">Services</a></li>
17                     <li><a href="#" data-i18n="nav.contact">Contact</a></li>
18                 </ul>
19                 <div class="lang-switcher">
20                     <label for="lang-select" data-i18n="lang">Language:</label>
21                     <select id="lang-select" aria-label="Select language">
22                         <option value="en">English</option>
23                         <option value="es">Español</option>
24                         <option value="ar">العربية</option>
25                     </select>
26                 </div>
27             </div>
28         </nav>
29     </header>
30
31     <main>
32         <section class="hero">
33             <div class="hero-container">
34                 <h1 data-i18n="hero.heading">Discover Amazing Things</h1>
35                 <p data-i18n="hero.subheading">Explore our world-class services and products</p>
36                 <button data-i18n="hero.cta">Get Started</button>
37             </div>
38         </section>
39
40         <section class="about">
41             <div class="container">
42                 <h2 data-i18n="about.title">About Us</h2>
43                 <p data-i18n="about.content">We are a leading company providing innovative solutions</p>
44             </div>

```

```
46
47     <section class="services">
48         <div class="container">
49             <h2 data-i18n="services.title">Our Services</h2>
50             <div class="services-grid">
51                 <div class="service">
52                     <h3 data-i18n="services.web">Web Development</h3>
53                     <p>Custom web solutions tailored to your needs.</p>
54                 </div>
55                 <div class="service">
56                     <h3 data-i18n="services.design">Graphic Design</h3>
57                     <p>Creative designs that capture your brand essence.</p>
58                 </div>
59                 <div class="service">
60                     <h3 data-i18n="services.consulting">Consulting</h3>
61                     <p>Expert advice to drive your business forward.</p>
62                 </div>
63             </div>
64         </div>
65     </section>
66
67     <section class="contact">
68         <div class="container">
69             <h2 data-i18n="contact.title">Contact Us</h2>
70             <form>
71                 <div class="form-group">
72                     <label for="name" data-i18n="contact.name">Name</label>
73                     <input type="text" id="name" name="name" required aria-required="true">
74                 </div>
75                 <div class="form-group">
76                     <label for="email" data-i18n="contact.email">Email</label>
77                     <input type="email" id="email" name="email" required aria-required="true">
78                 </div>
79                 <div class="form-group">
80                     <label for="message" data-i18n="contact.message">Message</label>
81                     <textarea id="message" name="message" required aria-required="true">
82                 </div>
83                 <button type="submit" data-i18n="contact.send">Send</button>
84             </form>
85         </div>
86     </section>
87 </main>
```

Output:



## Task 6 – Online Examination System

Prompt: Create an online exam interface.

Requirements:

- Timer, questions, and submission.
- Prevent multiple submissions.
- Use AI to:
  - Manage exam state.
  - Improve exam UX.
  - Ensure accessibility.

Code:

```

1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4    <meta charset="UTF-8">
5    <meta name="viewport" content="width=device-width, initial-scale=1.0">
6    <title>Online Examination System</title>
7    <link rel="stylesheet" href="exam.css">
8  </head>
9  <body>
10   <div id="exam-container">
11     <header>
12       <h1>Online Exam</h1>
13       <div class="exam-info">
14         <div class="timer" id="timer">Time: 30:00</div>
15         <div class="progress">
16           <span id="current-question">1</span> / <span id="total-questions">5
17         </div>
18       </div>
19     </header>
20
21     <main>
22       <div class="question-container" id="question-container">
23         <div class="question" id="question">
24           <h2 id="question-text">Question text here</h2>
25           <div class="options" id="options">
26             <!-- Options will be dynamically added -->
27           </div>
28         </div>
29       </div>
30
31       <div class="navigation">
32         <button id="prev-btn" disabled>Previous</button>
33         <button id="next-btn">Next</button>
34         <button id="submit-btn" style="display: none;">Submit Exam</button>
35       </div>
36     </main>
37   </div>
38
39   <div id="results" style="display: none;">
40     <h2>Exam Completed</h2>
41     <p>Your score: <span id="score"></span></p>
42     <button id="retake-btn">Retake Exam</button>
43   </div>
44
45   <script src="exam.js"></script>

```

Output:

The screenshot displays a web application titled "Online Exam". At the top, there is a dark blue header bar. On the left side of this bar, the text "Time: 28:33" is shown in red. On the right side, "2 / 5" is displayed in white. Below the header, a green progress bar is visible. The main content area has a white background and contains the question: "Which planet is known as the Red Planet?". Below the question, there are four radio button options: Venus, Mars, Jupiter, and Saturn. The interface is framed by a blue border on the left and a purple border on the right.